PART 1 — SHORT SCRIPTS FOR ALL 11 WEBSITE SLIDES (pitch.html#1 to #11)

WEBSITE SLIDES #1 – #11

Matches `soulnassim.github.io/Dortmund-OFF-grid-air/pitch.html` slide by slide, followed by your Full Left-to-Right Demo

Slide #1 · Opening Hook — `Dortmund OFF-Grid Air`

pitch.html#1

🎬 **ON SCREEN: Slide #1 (3D Dortmund Hero Animation). Pause 2 seconds, smile, look at the jury.**

"Imagine ten autonomous drones flying over Dortmund right now — delivering urgent blood supplies between hospitals, or helping firefighters see a big city fire from above.

Normally, those drones rely on public 5G or the internet to stay connected. ***But what happens during a real crisis when cell towers go down, or an attacker targets the network?*** *(1s pause)*

That is exactly where our mission, Dortmund OFF-Grid Air, comes in: we have to keep all ten drones flying and connected back to HQ at the Reinoldikirche with **zero public internet and zero cell towers** — which means they have to build their own private network right in the sky!"

Slide #2 · The Problem — `Today's fleets rely on 2.4 GHz, 4G/5G & GPS. In a crisis, all three fail.`

pitch.html#2

🎬 **ON SCREEN: Slide #2. Press [→] 3 times to reveal `01 · PUBLIC SPECTRUM`, `02 · STAR TOPOLOGY`, and `03 · BLIND NAVIGATION`.**

"Why do commercial drones fail immediately in a crisis today? Because three things break at once:

• **First (`01 · PUBLIC SPECTRUM — 2.4 GHz & 4G/5G jam`):** Standard drones fly on public 2.4 GHz radio or public 4G/5G cell towers. A cheap radio jammer or power blackout wipes both out in seconds.

• **Second (`02 · STAR TOPOLOGY — No relay, no QoS`):** Every drone only talks straight to the ground and **cannot bounce signals through neighbor drones (no relay)** — so one tall building cuts it off! Even worse, there is **no traffic priority (no QoS)**, meaning heavy HD camera video clogs the weak link and chokes out tiny steering commands.

• **And third (`03 · BLIND NAVIGATION — GPS spoofed, alarms late`):** Standard autopilots blindly trust fake GPS coordinates, while standard rolling-average alarms stay blind to slow-ramping attacks for over **9 seconds**!"

Slide #3 · Our Answer — `Optical lasers beat the jammer. Now the city is the obstacle.`

pitch.html#3

🎬 **ON SCREEN: Slide #3 (U-Tower 95m diagram + `22–28 m`, `120 m`, `3 s` on the right). Press [→] 3 times.**

"**To defeat radio jammers completely, we move our primary connection onto invisible 1550 nm Free-Space Optical (FSOC) lasers!** Because a laser is a narrow beam of light — not a radio wave — a radio jammer on the ground cannot jam it!

However, lasers create a new challenge: Dortmund's 3D buildings. Look at the diagram on the left: while typical Dortmund blocks are **22 to 28 meters** tall, the **Dortmunder U-Tower reaches 95 meters**. When `DR-03` flies at **50 meters** behind the U-Tower, the building blocks the laser beam from `HQ (65 m)`!

Therefore, we place **two High-Altitude Relay Drones (`DR-01` and `DR-02`) at 120 meters** — orbiting above every roof in Dortmund — with a **3-second hysteresis** so links never flicker on building edges."

Slide #4 · Hybrid Mesh — `Two off-grid links per drone (FSOC 1550 nm + DECT NR+ 1.9 GHz)`

pitch.html#4

🎬 **ON SCREEN: Slide #4 (Two columns: FSOC Laser 1550 nm vs DECT NR+ 1.9 GHz + `POD SWaP: 225 g · 9.7 W`).**

"Every drone carries a lightweight **225-gram, 9.7-watt hybrid pod** (signed with HMAC-SHA256) running **two off-grid links that protect each other**:

• **On the left (Primary · FSOC Laser 1550 nm):** When Line-of-Sight is clear, drones talk at **1 Gigabit per second** and **2 ms latency** with **0% radio jamming loss** — though dense fog scatters laser light beyond 520 meters (Mie scattering).

• **On the right (Fallback · DECT NR+ 1.9 GHz):** ***What happens when a building or dense fog blocks the laser?*** It switches to **DECT NR+** — **the world's first license-free 5G-standard radio that lets drones build their own private network in the sky (with zero SIM cards and zero cell towers)**, diffracting around buildings and penetrating dense fog with **0 dB loss**!

• ***And what if the opposite happens — what if an attacker jams the radio?*** Then any drone with clear Line-of-Sight stays 100% immune on the green laser!"

Slide #5 · Layer 1: Jamming Physics — `Distance decides. We don't cheat it.`

pitch.html#5

🎬 **ON SCREEN: Slide #5 (Concentric circles diagram `OUTER ZONE · 620 m` & `CORE <150 m`). Press [→] 3 times.**

"Jamming power follows the inverse-square law ($1/r^2$), so distance decides what happens across three zones:

1. **Clear Line-of-Sight · Anywhere (Green):** The laser holds with **0% loss** because photons ignore radio noise.

2. **Behind a building · `>150 m` in the Outer Zone (Orange — `DR-07` at 455 m):** The building blocks the laser, so DR-07 uses **DECT NR+** and triggers **QoS Emergency mode**: heavy video drops to **0%** so **C2 steering stays locked at 12 kbps (100%)**!

3. **Behind a building · `<150 m` in the Core (Purple — `DR-03` at 53 m):** Both the laser and radio are blocked at once. We don't cheat physics — the drone has to solve it physically."

PART 1 (CONT.) — WEBSITE SLIDES (#6 to #11)

Slide #6 (Pop-Up) · Slide #7 (Contrastive 0.0s) · Slide #8 (LLM Co-Pilot) · Slide #9 (Results) · Slide #10 (Team) · Slide #11 (Demo)

WEBSITE SLIDES #6 – #11

Slide #6 · Autonomous Recovery — `When every radio fails, climb.`

pitch.html#6

🎬 **ON SCREEN: Slide #6 (Vertical Pop-Up diagram + 4 timeline steps: `0.0 s`, `1.5 s`, `>95 m`, `3.2 s`). Press [→] 4 times.**

"***So what does a drone do when every radio fails at `0.0 s`?*** *(1s pause)* **It climbs!**

• At **`1.5 s`**: After 1.5 seconds isolated behind the U-Tower, DR-03 switches to its onboard **internal movement and camera sensors (`INS/VIO` — needing zero GPS and zero radio)** and climbs straight up from 50 m to 125 m.

• At **`>95 m`** → **`3.2 s`**: The moment it clears the **95-meter U-Tower roof**, Line-of-Sight to **`Relay DR-01 (120 m)`** opens up, and in **3.2 seconds**, the green FSOC laser snaps right back online!"

Slide #7 · Layer 2: Threat Detection — `Catch the slow attack at onset (9.5s vs 0.0s)`

pitch.html#7

🎬 **ON SCREEN: Slide #7 (`Threshold rules 9.5s` vs `Contrastive prototypes 0.0s`). Press [→] to reveal all items.**

"Physics keeps the link alive — ***but how will tomorrow's smart attackers try to bypass standard threat detectors?*** *(1s pause)*

Instead of turning a jammer on at full power all at once, a smart attacker **increases jamming power very slowly (+1.55 dB/s) — a Ramping Jammer**. ***Why does that fool standard threshold rules for 9.5 seconds?*** Because a standard detector only compares the current signal to the **moving average of the last 4 seconds** — when jamming increases slowly, that 4-second average shifts down right along with the attack, just like your eyes adjusting to a slowly darkening room!

Our Contrastive Prototype Detector catches it at 0.0 seconds: at takeoff, it freezes a **healthy baseline** of 4 sensors (Signal Strength, Signal Clarity, Packet Loss, and GPS-to-INS offset). Because that takeoff baseline never shifts, it catches slow jamming and slow GPS drift at **0.0s** with **0 false flags**! And when a novel combined attack hits, it stays honest (MPI-SP principle) and flags **Unexplained** instead of guessing."

Slide #8 · Layer 3: LLM Co-Pilot — `The LLM never detects. It explains.`

pitch.html#8

🎬 **ON SCREEN: Slide #8 (4-step pipeline: `4D telemetry` → `Rules + contrastive detector` → `Verdict` → `LLM analyst`). Press [→] 4 times.**

"**And because an `unexplained` verdict needs a label, that is where our LLM Security Co-Pilot steps in!**

Look at our pipeline: **the LLM never flies the drone and never detects** — detection is deterministic math (`z-score` + `cosine similarity` on our 4D telemetry). The LLM sits strictly downstream **offline at `HQ`** (with university LLM support via one env variable): it writes a **two-sentence German briefing**, files a **`LOW / MED / CRIT` ticket**, and gives a **one-click `Confirm Threat` command**.

When the commander clicks `Confirm Threat` once, it triggers Collaborative Fleet Learning: HQ broadcasts that tiny **4-number threat fingerprint to all 10 drones** — so every approaching drone recognizes it at 0.0s and switches to its DECT backup route before its link breaks (keeping recovery under 3s) — **cutting fleet downtime, sinking operational costs, and keeping hospital deliveries on schedule for Morpheus Logistik!**"

Slides #9, #10 & #11 · Results (`10/10 · 3.2s · 0.0s · 225g`), Team & Launch Live Demo

pitch.html#9 – #11

🎬 **ON SCREEN: Slide #9 (KPIs + 3-Phase Roadmap) → Slide #10 (Team) → Slide #11 (Click Green Button `Open mission control → index.html`).**

• **Slide #9 (Results & F&E Roadmap):** "In numbers: **10/10 drones reachable** under jamming and dense fog, **3.2s worst-case recovery**, **0.0s slow-attack detection**, and a **225g pod** plugging into Morpheus Leitstand via MAVLink 2 — scaling from Phase 1 COTS DECT NR+ bench test (Months 1–3) to Phase 2 1550 nm gimbal flight test (Months 4–8) and Phase 3 10-drone BVLOS trial between Klinikum Mitte and Westfalenhallen (Months 9–12)."

• **Slide #10 (The Team):** "Built by **Nassim Soultana** (3D Aerospace, FSOC/DECT, Fog Physics & INS/VIO Pop-Up), **Ilias Amir Soultana** (Contrastive Prototype AI, HMAC-SHA256 & LLM Co-Pilot), and **Sami Soultana** (QoS Shedding, 225g SWaP & Morpheus Ops)."

• **Slide #11 (Live Demo):** "Now let's click **Open Mission Control** and walk through the live system from left to right!"

PART 2 — FULL LEFT-TO-RIGHT LIVE DEMO SCRIPT (index.html)

LIVE DASHBOARD SCRIPT

Your Complete Left-to-Right Walkthrough of Mission Control (Left Sidebar → Center 3D Map → Right Sidebar)

Demo Step 1 · Left Sidebar — The 10 Drones & Reciprocal FSOC + DECT NR+ Channels

Left Panel

"On the left side, you see our ten drones — `DR-01` through `DR-10` — each with its own private sky-network IP address (`10.41.0.1` to `10.41.0.10`) talking back to **HQ** at the Reinoldikirche:

• When the sky is clear, they use **green FSOC Lasers** (1 Gbps, 2–5 ms latency). When a building or **Dense Fog (`[W]`)** blocks the laser beyond 520 meters, they automatically switch to **orange DECT NR+** (1.9 GHz) bouncing up to our **120-meter relays (`DR-01` & `DR-02`)**. And if a jammer blocks radio, any drone with clear Line-of-Sight stays 100% immune on the green laser!"

Demo Step 2 · Center 3D Map — The 3 Attack Circles Around the U-Tower

Center 3D Map

🎬 **ACTION: Press [J] Jamming and [G] GPS Spoofing (or click buttons at the bottom bar).**

"In the center 3D map around the **95-meter Dortmunder U-Tower**, drones outside the circles (like `DR-04` and `DR-08`) stay green because a ground jammer has limited range. Inside the attack zone, you see three concentric rings:

1. **Outer Yellow Circle (GPS Spoofing):** Drones cross-check GPS against their **internal movement sensor (`INS`)**, reject the fake GPS jump, and hold their true flight path.

2. **Middle Red Circle (Radio Jamming — `DR-07`):** Switches to **DECT NR+ backup radio** and cuts heavy video to 0%.

3. **Inner Purple Circle (`DOUBLE-FAIL` — `DR-03`):** Behind the U-Tower at 50m, both laser and radio are blocked! After 1.5s, `DR-03` uses **INS/VIO** to **climb straight up (`POP-UP` to 125m)**, peeks over the 95m roof, and restores its green laser in **3.2 seconds**!"

Demo Step 3 · Right Sidebar (Top & Middle) — Resilience Metrics & QoS Traffic Shedding

Right Top & Mid

"At the top of the Right Sidebar, the **4 Resilience Metrics** show **100% Reachability**, **Average Latency**, **Packet Loss**, and **3.2s Recovery Time**. Right below, under **QoS Traffic Priorities** (`C2 Command`, `Telemetry`, `Inter-Drone` 3D anti-collision/laser aiming, and `Video`), because video eats **99% of bandwidth** while steering needs less than **1% (12 kbps)**, our system automatically sheds video to **0%** so **C2 Steering stays locked at 100%**!"

Demo Step 4 · Right Sidebar (Bottom) — Contrastive Detector, LLM Co-Pilot & 1-Click Fleet Learning

Right Bottom

🎬 **ACTION: Click [🎯 Coordinated RF+GNSS] (Turns YELLOW) → Click [✓ Confirm Threat] (Turns BLUE!).**

"At the bottom right, when I trigger **`Coordinated RF+GNSS`**, our **Contrastive Detector** flags `DR-03` and `DR-07` in yellow as **UNEXPLAINED — needs label**. Right below, our offline **LLM Security Analyst** translates the 4D sensor jump into two German sentences, opens a `CRIT` ticket, and gives me **`Confirm Threat`**. When I click **`Confirm Threat`** once, **Collaborative Fleet Learning** broadcasts that 4-number threat fingerprint from `HQ` to all 10 drones — turning them **blue** so approaching drones proactively switch to DECT backup routes (<3s recovery), cutting downtime and cost for Morpheus Logistik!"

Technical Deep-Dive Notes (For Jury Q&A Only)

Q&A READY

1. How Z-Score (>= 4.5) & Cosine Similarity (>= 0.84) + Detector-to-LLM JSON Work

• **Z-Score Gate:** 40s (80-tick) takeoff warmup freezes `(mean, std)` of `[RSSI, SNR, Loss, GNSS_INS_Offset]`. Euclidean length `z_score >= 4.5` triggers detection at `0.0s`. **Cosine Similarity:** Compares 4D Z-vector direction against known prototypes; `Coordinated RF+GNSS` scores `sim = 0.68 < 0.84` → `UNEXPLAINED`.

• **LLM JSON & 1-Click Learning:** Detector passes `{device, label, z_score, dims, similarity}` to `llm_analyst.py` at HQ. Clicking `Confirm Threat` stores the normalized 4-float (16-byte) unit vector in the shared fleet prototype table (`sim` jumps to `0.99` across all 10 drones). Latency & speed are excluded from the 4D detector so building handovers (5ms → 40ms) cause **0 false cyberattack alarms**.